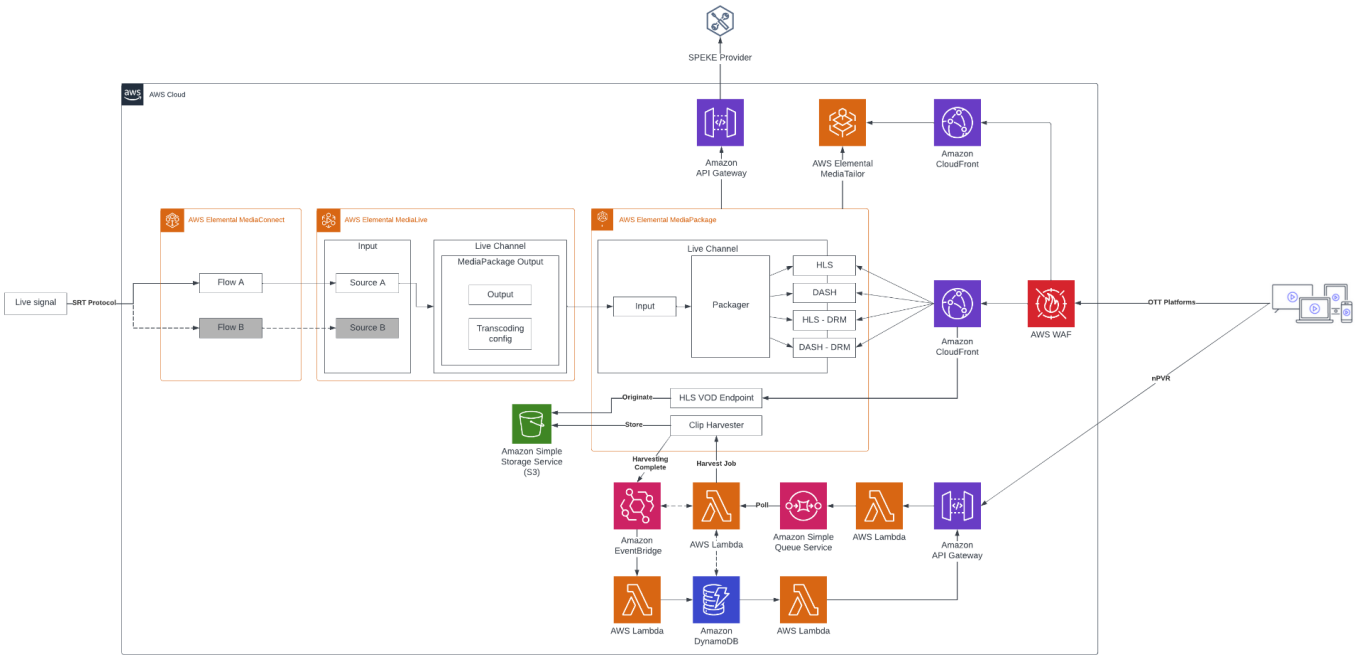
Live channels



The above diagram shows the general architecture to enable live channel delivery via AWS Elemental MediaConnect using the SRT protocol in listener mode.

Only one MediaConnect Flow is used, but in the two MediaConnect Flows can be used to implement a redundancy system/

AWS Elemental MediaLive takes one MediaConnect Flow as input (SINGLE PIPELINE). In the case where redundancy is required we could take two MediaConnect Flows as input (STANDARD CHANNEL).

This input is then attached to a MediaLive channel that produces an ABR output for AWS Elemental MediaPackage.

You will find below the MediaLive channel configuration (validated by Elemental team). FreeCast needed a fixed number of output and being cost efficient:

```
{
  "Arn": "REMOVED_SENSIBLE",
  "ChannelClass": "SINGLE_PIPELINE",
  "Destinations": [
```

```
    {
      "Id": "MediaPackage",
      "MediaPackageSettings": [
        {
          "ChannelId": "vubiquity_Value_AXS_TV"
        }
      ],
      "Settings": []
    }
  ],
  "EgressEndpoints": [
    {
      "SourceIp": "REMOVED_SENSIBLE"
    }
  ],
  "EncoderSettings": {
    "AudioDescriptions": [
      {
        "AudioTypeControl": "FOLLOW_INPUT",
        "CodecSettings": {
          "AacSettings": {
            "Bitrate": 64000,
            "CodingMode": "CODING_MODE_2_0",
            "InputType": "NORMAL",
            "Profile": "LC",
            "RateControlMode": "CBR",
            "RawFormat": "NONE",
            "SampleRate": 48000,
            "Spec": "MPEG4"
          }
        },
        "LanguageCodeControl": "FOLLOW_INPUT",
        "Name": "audio_aac64"
      },
      {
        "AudioTypeControl": "FOLLOW_INPUT",
        "CodecSettings": {
          "AacSettings": {
            "Bitrate": 128000,
            "CodingMode": "CODING_MODE_2_0",
            "InputType": "NORMAL",
            "Profile": "LC",
            "RateControlMode": "CBR",
            "RawFormat": "NONE",
            "SampleRate": 48000,
```

```
          "Spec": "MPEG4"
        }
      },
      "LanguageCodeControl": "FOLLOW_INPUT",
      "Name": "audio_aac128"
    }
  ],
  "CaptionDescriptions": [
    {
      "CaptionSelectorName": "English",
      "DestinationSettings": {
        "WebvttDestinationSettings": {
          "StyleControl": "PASSTHROUGH"
        }
      },
      "LanguageCode": "eng",
      "LanguageDescription": "English",
      "Name": "English"
    }
  ],
  "OutputGroups": [
    {
      "Name": "MediaPackage",
      "OutputGroupSettings": {
        "MediaPackageGroupSettings": {
          "Destination": {
            "DestinationRefId": "MediaPackage"
          }
        }
      },
      "Outputs": [
        {
          "AudioDescriptionNames": [
            "audio_aac64"
          ],
          "CaptionDescriptionNames": [],
          "OutputName": "360p",
          "OutputSettings": {
            "MediaPackageOutputSettings": {}
          },
          "VideoDescriptionName": "video_360p"
        },
        {
          "AudioDescriptionNames": [
            "audio_aac64"
```

```
          ],
          "CaptionDescriptionNames": [],
          "OutputName": "576p",
          "OutputSettings": {
            "MediaPackageOutputSettings": {}
          },
          "VideoDescriptionName": "video_576"
        },
        {
          "AudioDescriptionNames": [
            "audio_aac128"
          ],
          "CaptionDescriptionNames": [],
          "OutputName": "720p",
          "OutputSettings": {
            "MediaPackageOutputSettings": {}
          },
          "VideoDescriptionName": "video_720p"
        },
        {
          "AudioDescriptionNames": [
            "audio_aac128"
          ],
          "CaptionDescriptionNames": [],
          "OutputName": "1080p",
          "OutputSettings": {
            "MediaPackageOutputSettings": {}
          },
          "VideoDescriptionName": "video_1080p"
        },
        {
          "AudioDescriptionNames": [],
          "CaptionDescriptionNames": [
            "English"
          ],
          "OutputName": "English_WebVTT",
          "OutputSettings": {
            "MediaPackageOutputSettings": {}
          }
        }
      ]
    }
  ],
  "TimecodeConfig": {
    "Source": "SYSTEMCLOCK"
```

```
    },
    "VideoDescriptions": [
      {
        "CodecSettings": {
          "H264Settings": {
            "AdaptiveQuantization": "HIGH",
            "AfdSignaling": "NONE",
            "ColorMetadata": "INSERT",
            "EntropyEncoding": "CABAC",
            "FlickerAq": "ENABLED",
            "ForceFieldPictures": "DISABLED",
            "FramerateControl": "SPECIFIED",
            "FramerateDenominator": 1001,
            "FramerateNumerator": 30000,
            "GopBReference": "ENABLED",
            "GopClosedCadence": 1,
            "GopNumBFrames": 3,
            "GopSize": 60,
            "GopSizeUnits": "FRAMES",
            "Level": "H264_LEVEL_3_1",
            "LookAheadRateControl": "HIGH",
            "MaxBitrate": 900000,
            "NumRefFrames": 1,
            "ParControl": "SPECIFIED",
            "Profile": "MAIN",
            "QvbrQualityLevel": 6,
            "RateControlMode": "QVBR",
            "ScanType": "PROGRESSIVE",
            "SceneChangeDetect": "ENABLED",
            "SpatialAq": "ENABLED",
            "SubgopLength": "FIXED",
            "Syntax": "DEFAULT",
            "TemporalAq": "ENABLED",
            "TimecodeInsertion": "DISABLED"
          }
        },
        "Height": 360,
        "Name": "video_360p",
        "RespondToAfd": "NONE",
        "ScalingBehavior": "DEFAULT",
        "Sharpness": 50,
        "Width": 640
      },
      {
        "CodecSettings": {
```

```
        "H264Settings": {
          "AdaptiveQuantization": "HIGH",
          "AfdSignaling": "NONE",
          "ColorMetadata": "INSERT",
          "EntropyEncoding": "CABAC",
          "FlickerAq": "ENABLED",
          "ForceFieldPictures": "DISABLED",
          "FramerateControl": "SPECIFIED",
          "FramerateDenominator": 1001,
          "FramerateNumerator": 30000,
          "GopBReference": "ENABLED",
          "GopClosedCadence": 1,
          "GopNumBFrames": 3,
          "GopSize": 60,
          "GopSizeUnits": "FRAMES",
          "Level": "H264_LEVEL_4_1",
          "LookAheadRateControl": "HIGH",
          "MaxBitrate": 1500000,
          "NumRefFrames": 1,
          "ParControl": "SPECIFIED",
          "Profile": "MAIN",
          "QvbrQualityLevel": 7,
          "RateControlMode": "QVBR",
          "ScanType": "PROGRESSIVE",
          "SceneChangeDetect": "ENABLED",
          "SpatialAq": "ENABLED",
          "SubgopLength": "FIXED",
          "Syntax": "DEFAULT",
          "TemporalAq": "ENABLED",
          "TimecodeInsertion": "DISABLED"
        }
      },
      "Height": 576,
      "Name": "video_576",
      "RespondToAfd": "NONE",
      "ScalingBehavior": "DEFAULT",
      "Sharpness": 50,
      "Width": 1024
    },
    {
      "CodecSettings": {
        "H264Settings": {
          "AdaptiveQuantization": "HIGH",
          "AfdSignaling": "NONE",
          "ColorMetadata": "INSERT",
```

```
          "EntropyEncoding": "CABAC",
          "FlickerAq": "ENABLED",
          "ForceFieldPictures": "DISABLED",
          "FramerateControl": "SPECIFIED",
          "FramerateDenominator": 1001,
          "FramerateNumerator": 30000,
          "GopBReference": "ENABLED",
          "GopClosedCadence": 1,
          "GopNumBFrames": 3,
          "GopSize": 60,
          "GopSizeUnits": "FRAMES",
          "Level": "H264_LEVEL_4_1",
          "LookAheadRateControl": "HIGH",
          "MaxBitrate": 3500000,
          "NumRefFrames": 1,
          "ParControl": "SPECIFIED",
          "Profile": "MAIN",
          "QvbrQualityLevel": 8,
          "RateControlMode": "QVBR",
          "ScanType": "PROGRESSIVE",
          "SceneChangeDetect": "ENABLED",
          "SpatialAq": "ENABLED",
          "SubgopLength": "FIXED",
          "Syntax": "DEFAULT",
          "TemporalAq": "ENABLED",
          "TimecodeInsertion": "DISABLED"
        }
      },
      "Height": 720,
      "Name": "video_720p",
      "RespondToAfd": "NONE",
      "ScalingBehavior": "DEFAULT",
      "Sharpness": 50,
      "Width": 1280
    },
    {
      "CodecSettings": {
        "H264Settings": {
          "AdaptiveQuantization": "HIGH",
          "AfdSignaling": "NONE",
          "ColorMetadata": "INSERT",
          "EntropyEncoding": "CABAC",
          "FlickerAq": "ENABLED",
          "ForceFieldPictures": "DISABLED",
          "FramerateControl": "SPECIFIED",
```

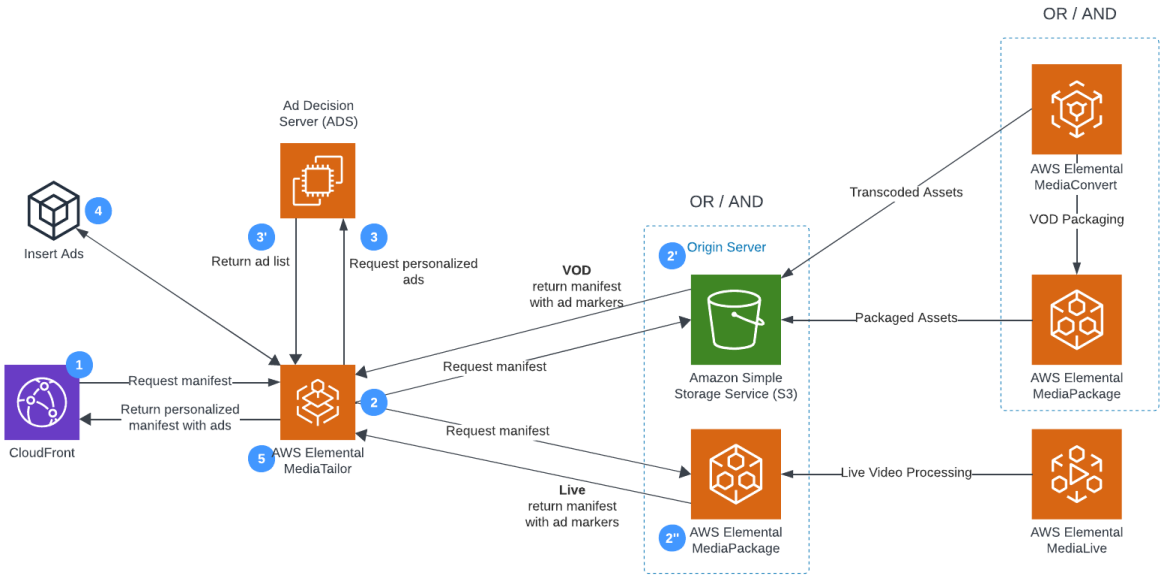
```
            "FramerateDenominator": 1001,
            "FramerateNumerator": 30000,
            "GopBReference": "DISABLED",
            "GopClosedCadence": 1,
            "GopNumBFrames": 1,
            "GopSize": 60,
            "GopSizeUnits": "FRAMES",
            "Level": "H264_LEVEL_4_1",
            "LookAheadRateControl": "HIGH",
            "MaxBitrate": 5000000,
            "NumRefFrames": 1,
            "ParControl": "SPECIFIED",
            "Profile": "MAIN",
            "QvbrQualityLevel": 9,
            "RateControlMode": "QVBR",
            "ScanType": "PROGRESSIVE",
            "SceneChangeDetect": "ENABLED",
            "SpatialAq": "ENABLED",
            "SubgopLength": "FIXED",
            "Syntax": "DEFAULT",
            "TemporalAq": "ENABLED",
            "TimecodeInsertion": "DISABLED"
          }
        },
        "Height": 1080,
        "Name": "video_1080p",
        "RespondToAfd": "NONE",
        "ScalingBehavior": "DEFAULT",
        "Sharpness": 50,
        "Width": 1920
      }
    ]
  },
  "Id": "9659842",
  "InputAttachments": [
    {
      "InputId": "278465",
      "InputSettings": {
        "AudioSelectors": [],
        "CaptionSelectors": [
          {
            "Name": "English",
            "SelectorSettings": {
              "EmbeddedSourceSettings": {
                "Convert608To708": "DISABLED",
```

```
                "Scte20Detection": "OFF",
                "Source608ChannelNumber": 1,
                "Source608TrackNumber": 1
              }
            }
          }
        ]
      }
    }
  ],
  "InputSpecification": {
    "Codec": "AVC",
    "MaximumBitrate": "MAX_10_MBPS",
    "Resolution": "HD"
  },
  "LogLevel": "DISABLED",
  "Maintenance": {
    "MaintenanceDay": "MONDAY",
    "MaintenanceStartTime": "11:00"
  },
  "MaintenanceStatus": "",
  "MaintenanceWindow": "MONDAY_1100",
  "Name": "vubiquity_Value_AXS_TV",
  "PipelineDetails": [],
  "PipelinesRunningCount": 0,
  "RoleArn": "REMOVED_SENSIBLE",
  "State": "IDLE",
  "Tags": {
    "map-migrated": "d-server-REMOVED_SENSIBLE",
    "aws:cloudformation:stack-name": "live-pipeline-prod",
    "aws:cloudformation:logical-id": "MediaLiveChannel27",
    "project": "freecast-live",
    "aws:cloudformation:stack-id": "REMOVED_SENSIBLE",
    "STAGE": "prod"
  }
}
```


MediaPackage then packages our MediaLive input to provide 4 types of endpoints:

- HLS
- HLS with DRM
- DASH
- DASH with DRM
-

An Gateway API has been set up to communicate with the external DRM provider and its SPEKE server.



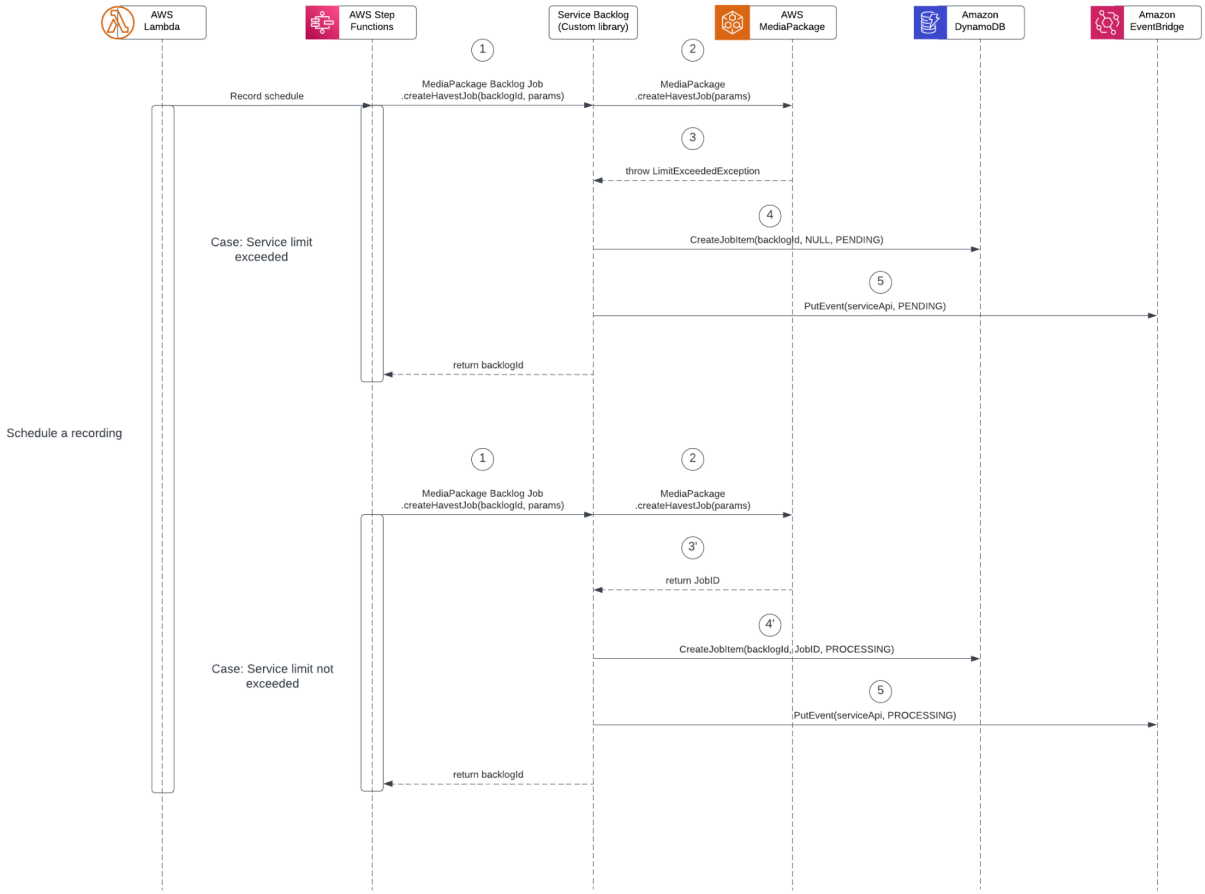
MediaTailor uses the HLS (non-encrypted) endpoint to perform ad insertion.
Two CloudFront Distributions will then be responsible for distributing the content to end users.

One distribution is dedicated to MediaTailor and another one is dedicated to the endpoints created by MediaPackage.

Firewall rules are also applied to the CloudFront distributions thanks to AWS WAF.
Another API Gateway is in charge of managing the CMS data.

It stores the metadata of the strings in an Amazon DynamoDB single table.

A queue system also allows users to schedule the recording of content to produce live-to-VOD (nPVR).



The system then uses MediaPackage's Harvest Job functionality to extract the content and store it in an Amazon S3 bucket before making it available to end users.

VOD (MRSS) Ingestion

An MRSS is a Media RSS, FreeCast has multiple different providers with different MRSS format.

i.e.:

```
<rss xmlns:media="http://search.yahoo.com/mrss/"
xmlns:xumo="http://www.xumo.com/rss/extensions/"
xmlns:dcterms="http://purl.org/dc/terms/" version="2.0">
<channel>
<title>axstv Content Catalog</title>
<language>en-us</language>
<item>
<guid isPermaLink="false">5eece84510134a335fd59e97</guid>
<title>Rock & Roll Road Trip with Sammy Hagar</title>
<description>Ride shotgun with Sammy as he laughs and jams with famed
rockers across the US.</description>
<media:thumbnail width="1280" height="720"
url="https://f9q4g5j6.ssl.hwcdn.net/5f2dc5b567d01c2c2112ddfa/1280/720"/>
<media:keywords>Music</media:keywords>
<xumo:episodic type="series"/>
</item>
<item>
<guid isPermaLink="false">5ef4ce5d9f4f26791af49252</guid>
<title>TrunkFest with Eddie Trunk</title>
<description>Explore the bands, cuisine, and attendees at the biggest music
events in the US.</description>
<media:thumbnail width="1280" height="720"
url="https://f9q4g5j6.ssl.hwcdn.net/5f2dc4efe486d766af187fb2/1280/720"/>
<media:keywords>TrunkFest with Eddie Trunk</media:keywords>
<xumo:episodic type="series"/>
</item>
<item>
<guid isPermaLink="false">5ef4ce8c7268db45822cbc42</guid>
<title>Drinking Made Easy</title>
<description>Search for the best drinks, pubs, and brews in this refreshing
travel series.</description>
<media:thumbnail width="1280" height="720"
url="https://f9q4g5j6.ssl.hwcdn.net/5f2dc837e486d766af187fb4/1280/720"/>
<media:keywords>Special</media:keywords>
<xumo:episodic type="series"/>
```

```
</item>
...
<item>
<title>Tool'n Around in Flavortown</title>
<description>This week chef and good friend, Guy Fieri, joins Sammy on his private jet for a jam at 20,000 feet. After catching up, they head to Jerome, Arizona to visit Maynard Keenan of Tool's one-of-a-kind winery to get a taste of his passion for food, wine and music.</description>
<guid isPermalink="false">5f78136eb70e4fcc60693eb8</guid>
<media:group>
<media:content duration="1276" type="video/mp4"
url="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2/assets/videos/5f78136eb70e4fcc60693eb8/vod/5f78136eb70e4fcc60693eb8.mp4"/>
<media:content duration="1276" type="application/x-mpegURL"
url="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2/assets/videos/5f78136eb70e4fcc60693eb8/vod/5f78136eb70e4fcc60693eb8.m3u8"/>
</media:group>
<media:content>
<media:thumbnail width="1280" height="720"
url="https://images.dotstudiopro.com/5f7ba43bb8bc7e7c9e054923/1280/720"/>
<media:thumbnail width="480" height="720"
url="https://images.dotstudiopro.com/5f7ba41c204db91ca6742cb5/480/720"/>
</media:content>
<xumo:cuePoints>0.0,477.0,672.0,1035.0</xumo:cuePoints>
<media:keywords>Rock & Roll Road Trip with Sammy Hagar, Rock & Roll Road Trip with Sammy Hagar Season 4, Rock & Roll Road Trip with Sammy Hagar Episode 9,september 2022,february 2023,may 2023</media:keywords>
<media:subTitle type="text/vtt" lang="en-us"
href="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2/assets/videos/5f78136eb70e4fcc60693eb8/subtitles/5f78136eb70e4fcc60693eb8-English_1617644002.vtt"/>
<pubDate>2020-10-03T06:00:14.838Z</pubDate>
<dcterms:valid>start=2018-07-08T00:00:00.000+00:00;end=2038-01-19T03:14:07.000+00:00;</dcterms:valid>
<media:rating scheme="urn:v-chip">tv-14</media:rating>
<media:category scheme="http://www.xumo.com">Music</media:category>
<xumo:episodic type="episode">
<xumo:seriesId>5eece84510134a335fd59e97</xumo:seriesId>
<xumo:episodeNum>9</xumo:episodeNum>
<xumo:seasonNum>4</xumo:seasonNum>
</xumo:episodic>
</item>
<item>
```

```
<title>Guitar Legends 2</title>
<description>Sammy sits down with Robby Krieger of The Doors and legendary
bluesman Joe Bonamassa at the second annual Guitar Legends: America Salutes
You event. Sammy and a host of legends take the stage for a good
cause.</description>
<guid isPermalink="false">5f78136fb70e4fcc60693eb9</guid>
<media:group>
<media:content duration="1282" type="video/mp4"
url="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2/
assets/videos/5f78136fb70e4fcc60693eb9/vod/5f78136fb70e4fcc60693eb9.mp4"/>
<media:content duration="1282" type="application/x-mpegURL"
url="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2/
assets/videos/5f78136fb70e4fcc60693eb9/vod/5f78136fb70e4fcc60693eb9.m3u8"/>
</media:group>
<media:content>
<media:thumbnail width="1280" height="720"
url="https://images.dotstudiopro.com/5f7baab4496e252dab5b0f38/1280/720"/>
<media:thumbnail width="480" height="720"
url="https://images.dotstudiopro.com/5f7baa9540cbe468434134b2/480/720"/>
</media:content>
<xumo:cuePoints>0.0,340.0,669.0,1016.0</xumo:cuePoints>
<media:keywords>Rock & Roll Road Trip with Sammy Hagar, Rock & Roll Road
Trip with Sammy Hagar Season 4, Rock & Roll Road Trip with Sammy Hagar
Episode 10,september 2022,february 2023,may 2023</media:keywords>
<media:subTitle type="text/vtt" lang="en-us"
href="https://b3u7d4z4.ssl.hwcdn.net/files/company/5e98e1f8bb2ee152df0229b2
/assets/videos/5f78136fb70e4fcc60693eb9/subtitles/5f78136fb70e4fcc60693eb9-
English_1617644002.vtt"/>
<pubDate>2020-10-03T06:00:15.226Z</pubDate>
<dcterms:valid>start=2018-07-15T00:00:00.000+00:00;end=2038-01-19T03:14:07.
000+00:00;</dcterms:valid>
<media:rating scheme="urn:v-chip">tv-14</media:rating>
<media:category scheme="http://www.xumo.com">Music</media:category>
<xumo:episodic type="episode">
<xumo:seriesId>5eece84510134a335fd59e97</xumo:seriesId>
<xumo:episodeNum>10</xumo:episodeNum>
<xumo:seasonNum>4</xumo:seasonNum>
</xumo:episodic>
</item>
</channel>
</rss>
```

In order to support different provider we use a mapping template per MRSS, this mapping

template supports XPath 3.0 expression

i.e.:

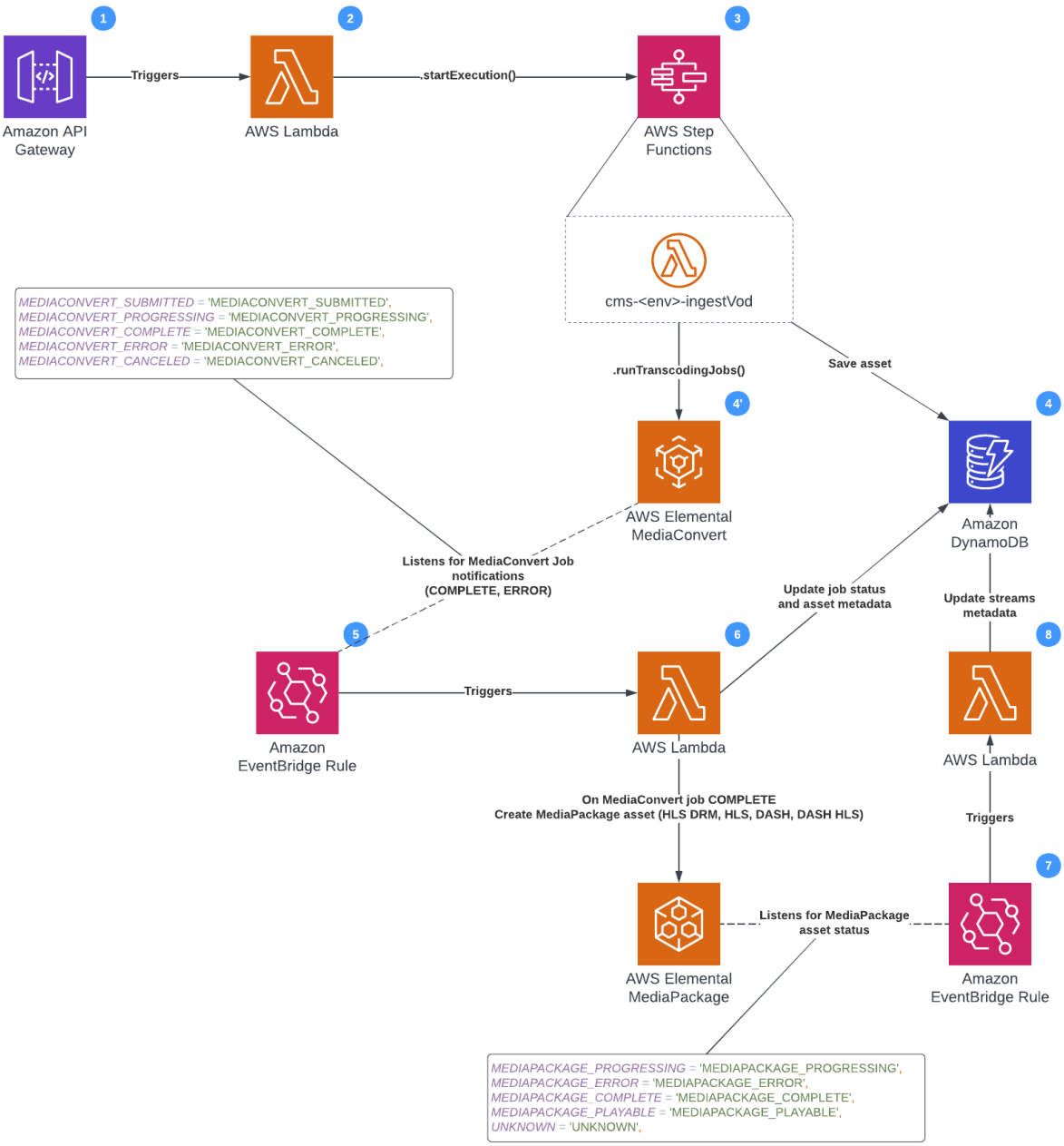
```
{
   "root":"//channel/item",
   "nextPage":"",
   "guid":"guid",
   "extId":"guid",
   "title":"title",
"asset":"*[name()='media:group']/*[name()='media:content'][@type='video/mp4
'] ! map:merge((map:entry('height', 'NA'), map:entry('width', 'NA'),
(@duration|@type|@url) ! map{local-name(): string()}))",
   "description":"if (exists(description) and string-length(description) >
0) then description else string(\"No description\")",
   "thumbnail":"array
{(*[name()='media:content']/*[name()='media:thumbnail'] |
*[name()='media:thumbnail']) ! map:merge((@height|@width|@url) !
map{local-name(): string()})}",
   "subtitle":"array {*[name()='media:subTitle'] !
map:merge((@type|@lang|@href) ! map{local-name(): string()})}",
   "pubDate":"if (exists(pubDate) and string-length(pubDate) > 0) then
pubDate else fn:current-date()",

"seasonNum":"*[name()='xumo:episodic']/*[name()='xumo:seasonNum']/text()",

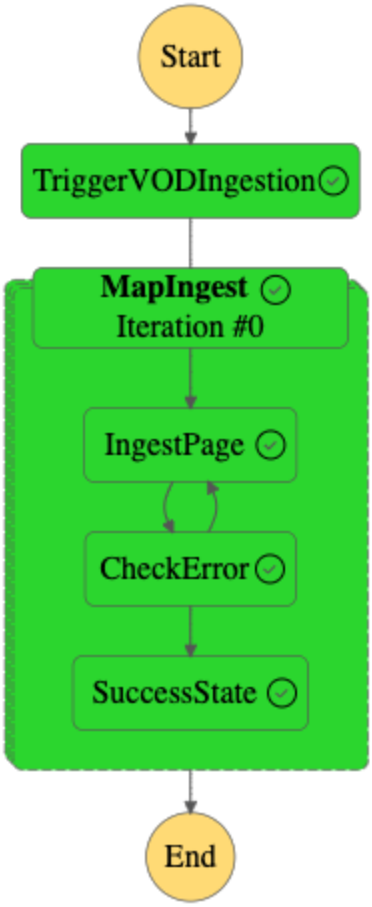
"episodeNum":"*[name()='xumo:episodic']/*[name()='xumo:episodeNum']/text()"
,
   "type":"*[name()='xumo:episodic']/@type",
   "seriesId":"*[name()='xumo:episodic']/*[name()='xumo:seriesId']/text()",
   "rotationStart":"if (exists(dcterms:valid) and
string-length(dcterms:valid) > 0 and contains(dcterms:valid, 'start='))
then (if (contains(dcterms:valid, ';')) then
substring-before(substring-after(translate(dcterms:valid, ' ', ''),
'start='), ';') else substring-after(translate(dcterms:valid, ' ', ''),
'start=')) else (fn:current-date())",
   "rotationEnd":"if (exists(dcterms:valid) and
string-length(dcterms:valid) > 0 and contains(dcterms:valid, 'end=')) then
(if (contains(dcterms:valid, ';')) then
substring-before(substring-after(translate(dcterms:valid, ' ', ''),
'end='), ';') else substring-after(translate(dcterms:valid, ' ', ''),
'end=')) else (fn:current-date())",
   "rating":"*[name()='media:rating']/text()",
   "cast":"",
```

```
   "genres":"if (*[name()='xumo:episodic'][@type='series']) then
*[name()='media:keywords']/text() else *[name()='media:category']/text()",
   "cuePoints":"*[name()='xumo:cuePoints']/text()"
}
```

This template is read by the parsing library we wrote. When a new ingestion is created (via API Gateway) it triggers a Lambda function that start an async workflow using Step Functions:



```
ingestVodState:
 name: ${self:service}-${sls:stage}-vodStateMachine
 definition:
   Comment: "Lambda to trigger VOD ingestion process"
   StartAt: TriggerVODIngestion
   States:
     TriggerVODIngestion:
       Type: Task
       Resource:
         Fn::GetAtt: [ingestVod, Arn]
       Next: MapIngest
     MapIngest:
       Type: Map
       ItemsPath: "$.url"
       ResultPath: "$.url"
       ItemSelector:
         "type.$": "$.type"
         "url.$": "$$.Map.Item.Value"
         "mapping.$": "$.mapping"
         "ingestionParentId.$": "$.ingestionParentId"
         "isPrivate.$": "$.isPrivate"
         "provider.$": "$.provider"
         "packages.$": "$.packages"
       ItemProcessor:
         ProcessorConfig:
           Mode: INLINE
         StartAt: IngestPage
         States:
           IngestPage:
             Type: Task
             Resource:
               Fn::GetAtt: [ingestVod, Arn]
             Next: CheckError
           CheckError:
             Type: Choice
             Choices:
               - Variable: "$.type"
                 StringEquals: "FAILURE"
                 Next: IngestPage
               - Variable: "$.type"
                 StringEquals: "DONE"
                 Next: SuccessState
           SuccessState:
             Type: Succeed
       End: true
```



The first step checks for the number of page on the ingested MRSS (sometimes you can have up to 460 MRSS pages) then it iterate through each pages and create different entities within DynamoDB if it meets different mandatory conditions, for instance mandatory fields, cue points for Ad Insertion with MediaConvert of Closed Captions and so on.

If an item doesn't meet the condition, logs are stored within DynamoDB explaining the issue with the item and then it continues to the next item until it reaches the end of the map.

When an asset/item is valid and stored within DynamoDB, a MediaConvert job is created and added to a reserved queue. This queue has 12 RTS (Reserved Transcoded Slots). When a job is done, another one is automatically picked by MediaConvert, FreeCast didn't want to use the Standard queue at all.

If you ingest the same assets, the Lambda within the state machine will check the guid of the asset, if it's present within DynamoDB and correctly transcoded, the pipeline will ignore the asset and move to the next one.

MRSS Ingestion can be done via the CMS or the API

Here is the MediaConvert Job Template:

```
{
 "UserMetadata": {
   "entityId": "[Populated by MediaConvertTranscoder]"
 },
 "Role": "[Populated by MediaConvertTranscoder]",
 "Settings": {
   "OutputGroups": [
     {
       "CustomName": "hls_out",
       "Name": "Apple HLS",
       "Outputs": [
         {
           "ContainerSettings": {
             "Container": "M3U8",
             "M3u8Settings": {
               "AudioFramesPerPes": 4,
               "PcrControl": "PCR_EVERY_PES_PACKET",
               "PmtPid": 480,
               "PrivateMetadataPid": 503,
               "ProgramNumber": 1,
               "PatInterval": 0,
               "PmtInterval": 0,
               "Scte35Source": "PASSTHROUGH",
               "NielsenId3": "NONE",
               "TimedMetadata": "NONE",
               "VideoPid": 481,
               "AudioPids": [
                 482,
                 483,
                 484,
                 485,
                 486,
                 487,
                 488,
                 489,
                 490,
                 491,
                 492
               ]
             }
           },
           "VideoDescription": {
             "Width": 1920,
             "ScalingBehavior": "DEFAULT",
             "Height": 1080,
             "TimecodeInsertion": "DISABLED",
```

```
            "AntiAlias": "ENABLED",
            "Sharpness": 50,
            "CodecSettings": {
              "Codec": "H_264",
              "H264Settings": {
                "InterlaceMode": "PROGRESSIVE",
                "ParNumerator": 1,
                "NumberReferenceFrames": 3,
                "Syntax": "DEFAULT",
                "Softness": 0,
                "GopClosedCadence": 1,
                "HrdBufferInitialFillPercentage": 80,
                "GopSize": 2,
                "Slices": 1,
                "GopBReference": "DISABLED",
                "HrdBufferSize": 75000000,
                "MaxBitrate": 5000000,
                "SlowPal": "DISABLED",
                "ParDenominator": 1,
                "EntropyEncoding": "CABAC",
                "FramerateControl": "INITIALIZE_FROM_SOURCE",
                "RateControlMode": "QVBR",
                "CodecProfile": "MAIN",
                "Telecine": "NONE",
                "MinIInterval": 0,
                "AdaptiveQuantization": "HIGH",
                "CodecLevel": "LEVEL_4_1",
                "FieldEncoding": "PAFF",
                "SceneChangeDetect": "TRANSITION_DETECTION",
                "QualityTuningLevel": "SINGLE_PASS",
                "FramerateConversionAlgorithm": "DUPLICATE_DROP",
                "UnregisteredSeiTimecode": "DISABLED",
                "GopSizeUnits": "SECONDS",
                "ParControl": "SPECIFIED",
                "NumberBFramesBetweenReferenceFrames": 2,
                "RepeatPps": "DISABLED",
                "DynamicSubGop": "STATIC"
              }
            },
            "AfdSignaling": "NONE",
            "DropFrameTimecode": "ENABLED",
            "RespondToAfd": "NONE",
            "ColorMetadata": "INSERT"
          },
          "OutputSettings": {
            "HlsSettings": {
              "AudioGroupId": "program_audio",
```

```
            "IFrameOnlyManifest": "EXCLUDE",
            "AudioOnlyContainer": "AUTOMATIC"
          }
        },
        "NameModifier": "_1080p"
      },
      {
        "ContainerSettings": {
          "Container": "M3U8",
          "M3u8Settings": {
            "AudioFramesPerPes": 4,
            "PcrControl": "PCR_EVERY_PES_PACKET",
            "PmtPid": 480,
            "PrivateMetadataPid": 503,
            "ProgramNumber": 1,
            "PatInterval": 0,
            "PmtInterval": 0,
            "Scte35Source": "PASSTHROUGH",
            "NielsenId3": "NONE",
            "TimedMetadata": "NONE",
            "TimedMetadataPid": 502,
            "VideoPid": 481,
            "AudioPids": [
              482,
              483,
              484,
              485,
              486,
              487,
              488,
              489,
              490,
              491,
              492
            ]
          }
        },
        "VideoDescription": {
          "Width": 1280,
          "ScalingBehavior": "DEFAULT",
          "Height": 720,
          "TimecodeInsertion": "DISABLED",
          "AntiAlias": "ENABLED",
          "Sharpness": 50,
          "CodecSettings": {
            "Codec": "H_264",
            "H264Settings": {
```

```
            "InterlaceMode": "PROGRESSIVE",
            "ParNumerator": 1,
            "NumberReferenceFrames": 3,
            "Syntax": "DEFAULT",
            "Softness": 0,
            "GopClosedCadence": 1,
            "HrdBufferInitialFillPercentage": 80,
            "GopSize": 2,
            "Slices": 1,
            "GopBReference": "DISABLED",
            "HrdBufferSize": 75000000,
            "MaxBitrate": 3500000,
            "SlowPal": "DISABLED",
            "ParDenominator": 1,
            "EntropyEncoding": "CABAC",
            "FramerateControl": "INITIALIZE_FROM_SOURCE",
            "RateControlMode": "QVBR",
            "CodecProfile": "MAIN",
            "Telecine": "NONE",
            "MinIInterval": 0,
            "AdaptiveQuantization": "HIGH",
            "CodecLevel": "LEVEL_4_1",
            "FieldEncoding": "PAFF",
            "SceneChangeDetect": "TRANSITION_DETECTION",
            "QualityTuningLevel": "SINGLE_PASS",
            "FramerateConversionAlgorithm": "DUPLICATE_DROP",
            "UnregisteredSeiTimecode": "DISABLED",
            "GopSizeUnits": "SECONDS",
            "ParControl": "SPECIFIED",
            "NumberBFramesBetweenReferenceFrames": 2,
            "RepeatPps": "DISABLED",
            "DynamicSubGop": "STATIC"
          }
        },
        "AfdSignaling": "NONE",
        "DropFrameTimecode": "ENABLED",
        "RespondToAfd": "NONE",
        "ColorMetadata": "INSERT"
      },
      "OutputSettings": {
        "HlsSettings": {
          "AudioGroupId": "program_audio",
          "IFrameOnlyManifest": "EXCLUDE",
          "AudioOnlyContainer": "AUTOMATIC"
        }
      },
      "NameModifier": "_720p"
```

```
        },
        {
          "ContainerSettings": {
            "Container": "M3U8",
            "M3u8Settings": {
              "AudioFramesPerPes": 4,
              "PcrControl": "PCR_EVERY_PES_PACKET",
              "PmtPid": 480,
              "PrivateMetadataPid": 503,
              "ProgramNumber": 1,
              "PatInterval": 0,
              "PmtInterval": 0,
              "Scte35Source": "PASSTHROUGH",
              "NielsenId3": "NONE",
              "TimedMetadata": "NONE",
              "VideoPid": 481,
              "AudioPids": [
                482,
                483,
                484,
                485,
                486,
                487,
                488,
                489,
                490,
                491,
                492
              ]
            }
          },
          "VideoDescription": {
            "Width": 1024,
            "ScalingBehavior": "DEFAULT",
            "Height": 576,
            "TimecodeInsertion": "DISABLED",
            "AntiAlias": "ENABLED",
            "Sharpness": 50,
            "CodecSettings": {
              "Codec": "H_264",
              "H264Settings": {
                "InterlaceMode": "PROGRESSIVE",
                "ParNumerator": 1,
                "NumberReferenceFrames": 3,
                "Syntax": "DEFAULT",
                "Softness": 0,
                "GopClosedCadence": 1,
```

```
            "HrdBufferInitialFillPercentage": 80,
            "GopSize": 2,
            "Slices": 1,
            "GopBReference": "DISABLED",
            "HrdBufferSize": 21000000,
            "MaxBitrate": 1500000,
            "SlowPal": "DISABLED",
            "ParDenominator": 1,
            "EntropyEncoding": "CABAC",
            "FramerateControl": "INITIALIZE_FROM_SOURCE",
            "RateControlMode": "QVBR",
            "CodecProfile": "MAIN",
            "Telecine": "NONE",
            "MinIInterval": 0,
            "AdaptiveQuantization": "HIGH",
            "CodecLevel": "LEVEL_4_1",
            "FieldEncoding": "PAFF",
            "SceneChangeDetect": "TRANSITION_DETECTION",
            "QualityTuningLevel": "SINGLE_PASS",
            "FramerateConversionAlgorithm": "DUPLICATE_DROP",
            "UnregisteredSeiTimecode": "DISABLED",
            "GopSizeUnits": "SECONDS",
            "ParControl": "SPECIFIED",
            "NumberBFramesBetweenReferenceFrames": 2,
            "RepeatPps": "DISABLED",
            "DynamicSubGop": "STATIC"
          }
        },
        "AfdSignaling": "NONE",
        "DropFrameTimecode": "ENABLED",
        "RespondToAfd": "NONE",
        "ColorMetadata": "INSERT"
      },
      "OutputSettings": {
        "HlsSettings": {
          "AudioGroupId": "program_audio",
          "IFrameOnlyManifest": "EXCLUDE",
          "AudioOnlyContainer": "AUTOMATIC"
        }
      },
      "NameModifier": "_576p"
    },
    {
      "ContainerSettings": {
        "Container": "M3U8",
        "M3u8Settings": {
          "AudioFramesPerPes": 4,
```

```
            "PcrControl": "PCR_EVERY_PES_PACKET",
            "PmtPid": 480,
            "PrivateMetadataPid": 503,
            "ProgramNumber": 1,
            "PatInterval": 0,
            "PmtInterval": 0,
            "Scte35Source": "PASSTHROUGH",
            "NielsenId3": "NONE",
            "TimedMetadata": "NONE",
            "VideoPid": 481,
            "AudioPids": [
              482,
              483,
              484,
              485,
              486,
              487,
              488,
              489,
              490,
              491,
              492
            ]
          }
        },
        "VideoDescription": {
          "Width": 640,
          "ScalingBehavior": "DEFAULT",
          "Height": 360,
          "TimecodeInsertion": "DISABLED",
          "AntiAlias": "ENABLED",
          "Sharpness": 50,
          "CodecSettings": {
            "Codec": "H_264",
            "H264Settings": {
              "InterlaceMode": "PROGRESSIVE",
              "ParNumerator": 1,
              "NumberReferenceFrames": 3,
              "Syntax": "DEFAULT",
              "Softness": 0,
              "GopClosedCadence": 1,
              "HrdBufferInitialFillPercentage": 80,
              "GopSize": 2,
              "Slices": 1,
              "GopBReference": "DISABLED",
              "HrdBufferSize": 16800000,
              "MaxBitrate": 900000,
```

```
            "SlowPal": "DISABLED",
            "ParDenominator": 1,
            "EntropyEncoding": "CABAC",
            "FramerateControl": "INITIALIZE_FROM_SOURCE",
            "RateControlMode": "QVBR",
            "CodecProfile": "MAIN",
            "Telecine": "NONE",
            "MinIInterval": 0,
            "AdaptiveQuantization": "HIGH",
            "CodecLevel": "LEVEL_3_1",
            "FieldEncoding": "PAFF",
            "SceneChangeDetect": "TRANSITION_DETECTION",
            "QualityTuningLevel": "SINGLE_PASS",
            "FramerateConversionAlgorithm": "DUPLICATE_DROP",
            "UnregisteredSeiTimecode": "DISABLED",
            "GopSizeUnits": "SECONDS",
            "ParControl": "SPECIFIED",
            "NumberBFramesBetweenReferenceFrames": 2,
            "RepeatPps": "DISABLED",
            "DynamicSubGop": "STATIC"
          }
        },
        "AfdSignaling": "NONE",
        "DropFrameTimecode": "ENABLED",
        "RespondToAfd": "NONE",
        "ColorMetadata": "INSERT"
      },
      "OutputSettings": {
        "HlsSettings": {
          "AudioGroupId": "program_audio",
          "IFrameOnlyManifest": "EXCLUDE",
          "AudioOnlyContainer": "AUTOMATIC"
        }
      },
      "NameModifier": "_360p"
    },
    {
      "ContainerSettings": {
        "Container": "M3U8",
        "M3u8Settings": {
          "AudioFramesPerPes": 4,
          "PcrControl": "PCR_EVERY_PES_PACKET",
          "PmtPid": 480,
          "PrivateMetadataPid": 503,
          "ProgramNumber": 1,
          "PatInterval": 0,
          "PmtInterval": 0,
```

```
            "Scte35Source": "PASSTHROUGH",
            "NielsenId3": "NONE",
            "TimedMetadata": "NONE",
            "VideoPid": 481,
            "AudioPids": [
              482,
              483,
              484,
              485,
              486,
              487,
              488,
              489,
              490,
              491,
              492
            ]
          }
        },
        "AudioDescriptions": [
          {
            "AudioTypeControl": "FOLLOW_INPUT",
            "AudioSourceName": "Audio Selector 1",
            "CodecSettings": {
              "Codec": "AAC",
              "AacSettings": {
                "AudioDescriptionBroadcasterMix": "NORMAL",
                "Bitrate": 192000,
                "RateControlMode": "CBR",
                "CodecProfile": "LC",
                "CodingMode": "CODING_MODE_2_0",
                "RawFormat": "NONE",
                "SampleRate": 48000,
                "Specification": "MPEG4"
              }
            },
            "LanguageCodeControl": "FOLLOW_INPUT"
          }
        ],
        "OutputSettings": {
          "HlsSettings": {
            "AudioGroupId": "program_audio",
            "AudioTrackType": "ALTERNATE_AUDIO_AUTO_SELECT_DEFAULT",
            "IFrameOnlyManifest": "EXCLUDE",
            "AudioOnlyContainer": "AUTOMATIC"
          }
        },
```

```
          "NameModifier": "_audio"
        }
      ],
      "OutputGroupSettings": {
        "Type": "HLS_GROUP_SETTINGS",
        "HlsGroupSettings": {
          "ManifestDurationFormat": "INTEGER",
          "SegmentLength": 6,
          "TimedMetadataId3Period": 10,
          "CaptionLanguageSetting": "OMIT",
          "Destination": "[Populated by MediaConvertTranscoder]",
          "TimedMetadataId3Frame": "PRIV",
          "CodecSpecification": "RFC_4281",
          "OutputSelection": "MANIFESTS_AND_SEGMENTS",
          "ProgramDateTimePeriod": 600,
          "MinSegmentLength": 0,
          "MinFinalSegmentLength": 0,
          "DirectoryStructure": "SINGLE_DIRECTORY",
          "ProgramDateTime": "EXCLUDE",
          "SegmentControl": "SINGLE_FILE",
          "ManifestCompression": "NONE",
          "ClientCache": "ENABLED",
          "AudioOnlyHeader": "INCLUDE",
          "StreamInfResolution": "INCLUDE",
          "AdMarkers": [
            "ELEMENTAL"
          ]
        }
      }
    }
  ],
  "AdAvailOffset": 0,
  "Inputs": [
    {
      "AudioSelectors": {
        "Audio Selector 1": {
          "Offset": 0,
          "DefaultSelection": "DEFAULT",
          "ProgramSelection": 1
        }
      },
      "VideoSelector": {
        "ColorSpace": "FOLLOW",
        "Rotate": "DEGREE_0",
        "AlphaBehavior": "DISCARD"
      },
      "FilterEnable": "AUTO",
```

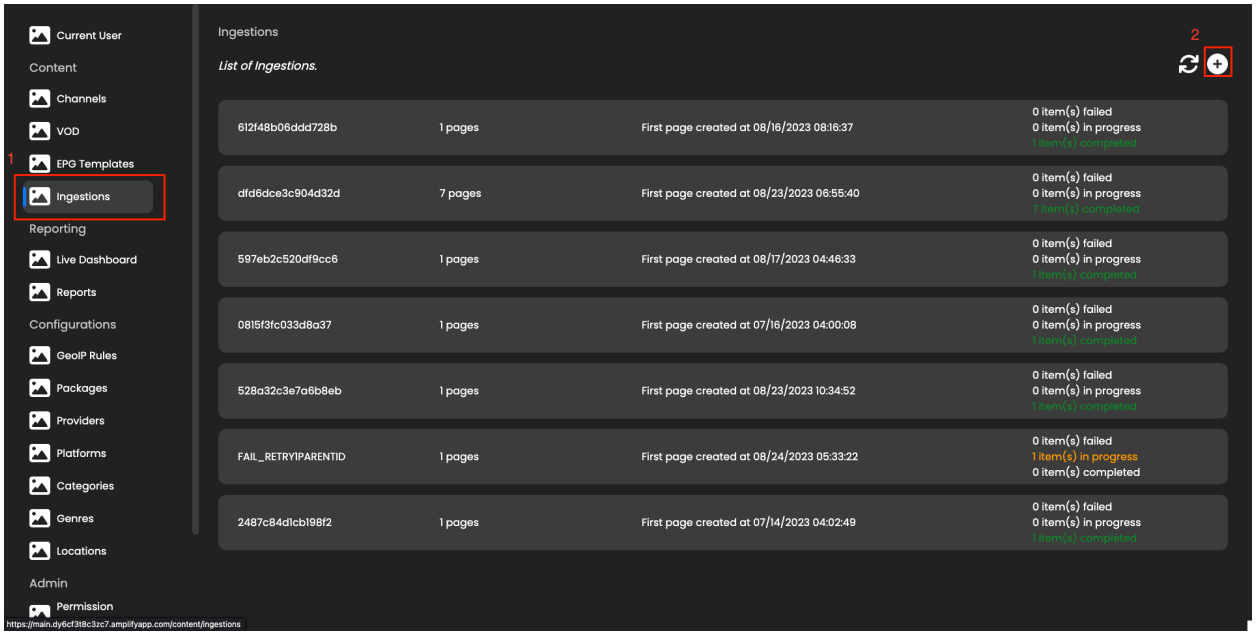
```
      "PsiControl": "USE_PSI",
      "FilterStrength": 0,
      "DeblockFilter": "DISABLED",
      "DenoiseFilter": "DISABLED",
      "InputScanType": "AUTO",
      "TimecodeSource": "ZEROBASED",
      "FileInput": "[Populated by MediaConvertTranscoder]"
    }
  ]
 },
 "AccelerationSettings": {
  "Mode": "DISABLED"
 },
 "StatusUpdateInterval": "SECONDS_60",
 "Priority": 0
}
```

Here is the MediaPackage Packaging Group configuration:

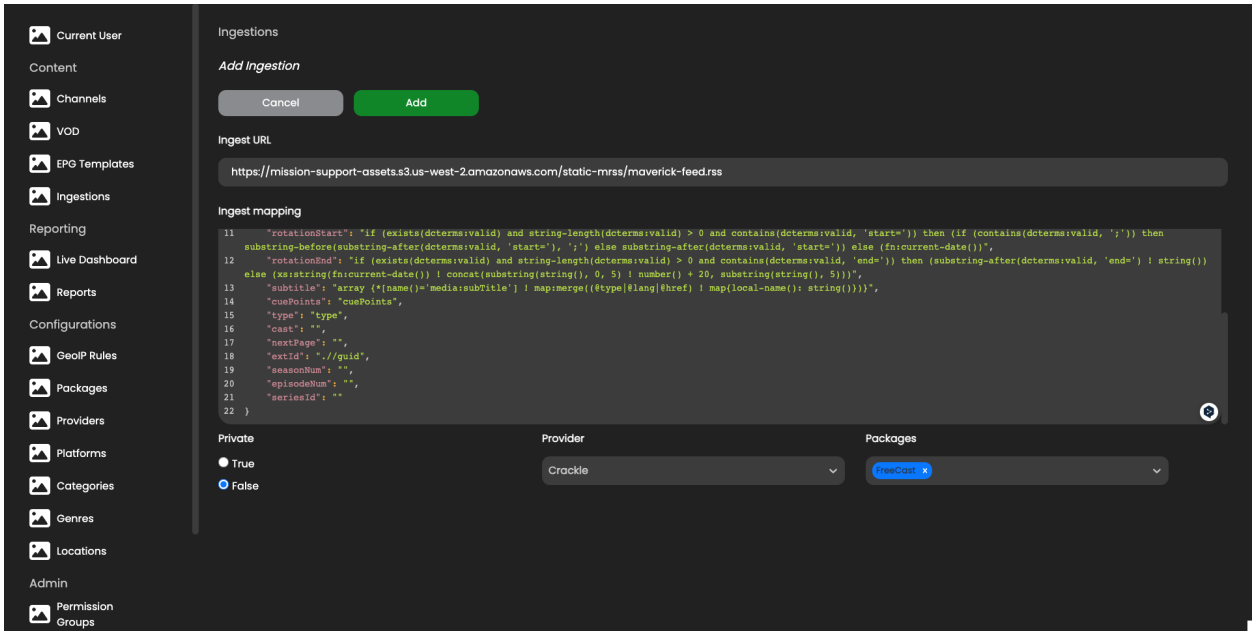
```
jobResources:
 packagingGroup:
   Type: AWS::MediaPackage::PackagingGroup
   Properties:
     Id: vod-${sls:stage}
     Authorization:
       CdnIdentifierSecret: !Ref mediaPackageCDNAuthorizationSecret
       SecretsRoleArn: !GetAtt MediaPackageASM.Arn
 hlsDrmPackagingConfiguration:
   Type: AWS::MediaPackage::PackagingConfiguration
   Properties:
     HlsPackage:
       Encryption:
         EncryptionMethod: 'SAMPLE_AES'
         SpekeKeyProvider:
           SystemIds:
             - '94ce86fb-07ff-4f43-adb8-93d2fa968ca2'
           Url: ${param:spekeUrl}/v1
           RoleArn: ${param:spekeRoleArn}
       HlsManifests:
         - AdMarkers: 'SCTE35_ENHANCED'
     Id: HLS_DRM_${sls:stage}
     PackagingGroupId: !Ref packagingGroup
 dashDrmPackagingConfiguration:
```

```
  Type: AWS::MediaPackage::PackagingConfiguration
  Properties:
    DashPackage:
      PeriodTriggers:
        - ADS
      DashManifests:
        - ManifestLayout: 'COMPACT'
      SegmentDurationSeconds: 6
      Encryption:
        SpekeKeyProvider:
          SystemIds:
            - 'edef8ba9-79d6-4ace-a3c8-27dcd51d21ed'
          Url: ${param:spekeUrl}/v1
          RoleArn: ${param:spekeRoleArn}
    Id: DASH_DRM_${sls:stage}
    PackagingGroupId: !Ref packagingGroup
hlsPackagingConfiguration:
  Type: AWS::MediaPackage::PackagingConfiguration
  Properties:
    HlsPackage:
      HlsManifests:
        - AdMarkers: 'SCTE35_ENHANCED'
    Id: HLS_${sls:stage}
    PackagingGroupId: !Ref packagingGroup
dashPackagingConfiguration:
  Type: AWS::MediaPackage::PackagingConfiguration
  Properties:
    DashPackage:
      PeriodTriggers:
        - ADS
      DashManifests:
        - ManifestLayout: 'COMPACT'
      SegmentDurationSeconds: 6
    Id: DASH_${sls:stage}
    PackagingGroupId: !Ref packagingGroup
```

CMS



1. Navigate to the Ingestions section
2. Click on the “plus” button at the top right corner



3. Provide:
    - The MRSS URL
    - The Mapping Template
    - The provider (if required)

- The packages (if required)

4. The “Private” attribute should be set to “True” if assets are located in a private S3 bucket. In order to allow AWS MediaConvert to access private assets within the S3 bucket, a policy needs to be added to the S3 bucket.

i.e.:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "StmtAllowMediaConvert131531546",
            "Effect": "Allow",
            "Principal": {
                "AWS": [

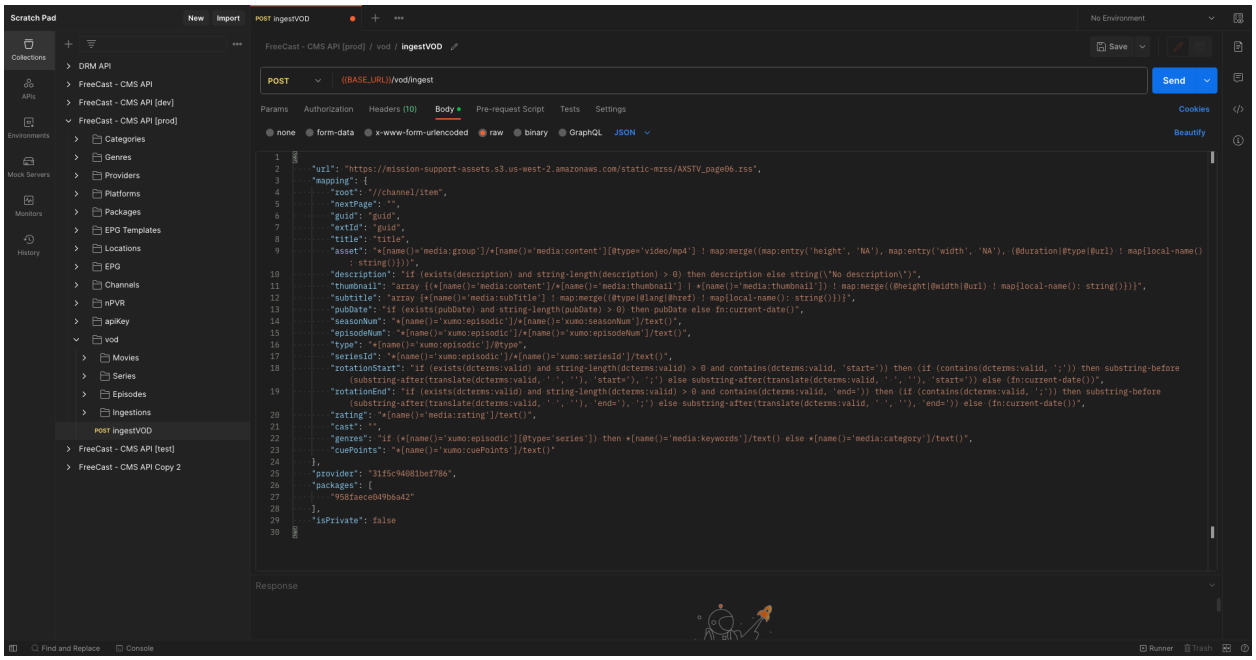
"arn:aws:iam::199718138082:role/cms-api-test-MediaConvertJobRole-B7SWC7FWVEL5",

"arn:aws:iam::199718138082:role/cms-api-dev-MediaConvertJobRole-1RCJRSEW1XQVF",

"arn:aws:iam::199718138082:role/cms-api-prod-MediaConvertJobRole-1OJRZXXQCZUNW"
                ]
            },
            "Action": [
                "s3:GetObject",
                "s3:ListBucket"
            ],
            "Resource": [
                "arn:aws:s3:::your_s3_bucket_arn",
                "arn:aws:s3:::your_s3_bucket_arn/*"
            ]
        }
    ]
}
```

API

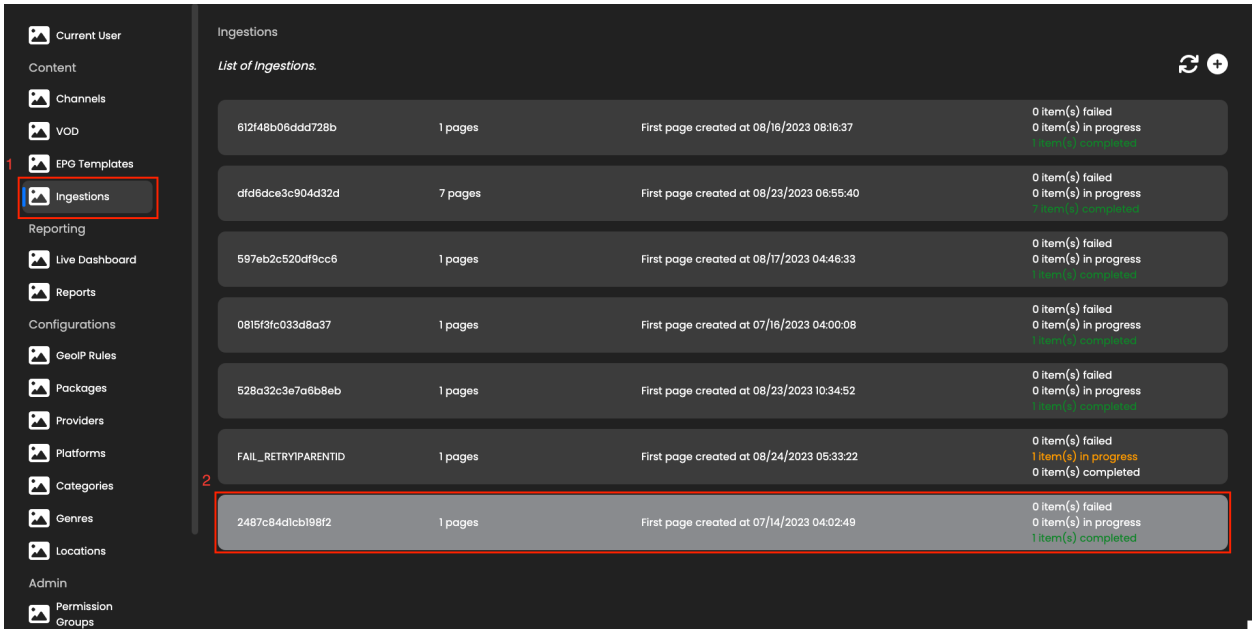
As said, you use the API as well to start a new MRSS ingestion, here we use Postman but cURL or any HTTP client can be used



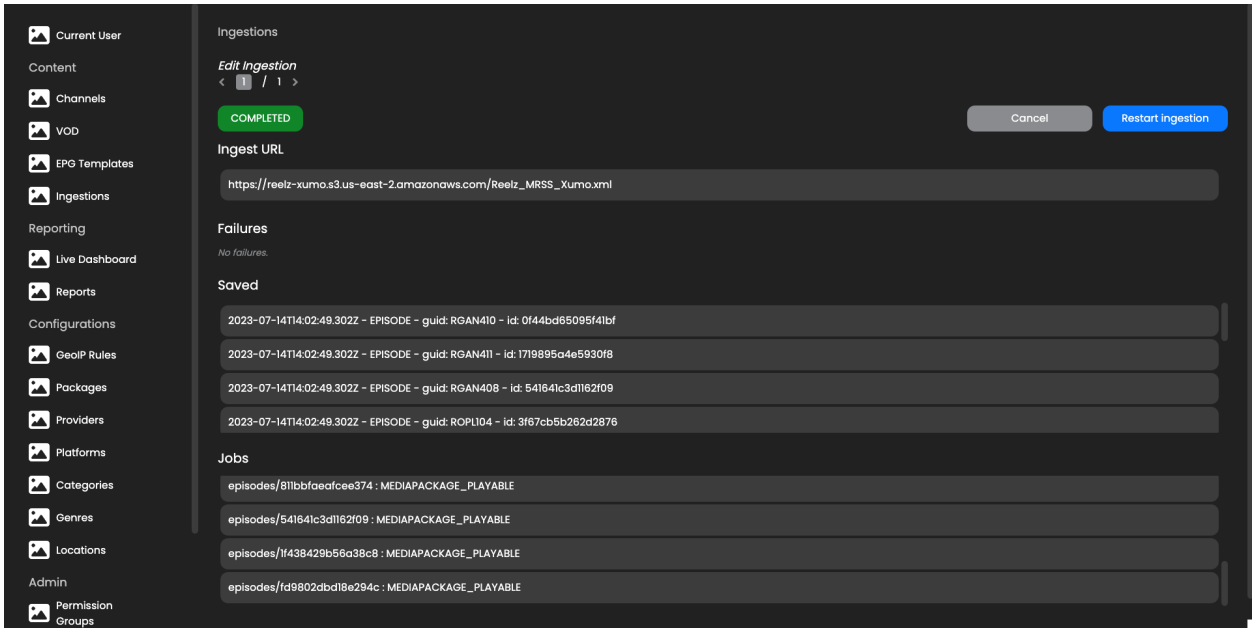

```
curl --location 'https://d1ja1lkih15vd5.cloudfront.net/prod/vod/ingest' \
--header 'x-api-key: API-KEY \
--header 'Content-Type: application/json' \
--data-raw '{
"url":
"https://mission-support-assets.s3.us-west-2.amazonaws.com/static-mrss/AXSTV_page06.rss",
"mapping": {
"root": "//channel/item",
"nextPage": "",
"guid": "guid",
"extId": "guid",
"title": "title",
"asset":
"*[name()='\''media:group'\'']/*[name()='\''media:content'\''][@type='\''video/mp4'\''] !
map:merge((map:entry('\''height'\'', '\''NA'\''), map:entry('\''width'\'', '\''NA'\''),
(@duration|@type|@url) ! map{local-name(): string()}))",
"description": "if (exists(description) and string-length(description) > 0) then description
else string(\"No description\")",
"thumbnail": "array {(*[name()='\''media:content'\'']/*[name()='\''media:thumbnail'\''] |
*[name()='\''media:thumbnail'\'']) ! map:merge((@height|@width|@url) ! map{local-name():
string()})}",
"subtitle": "array {*[name()='\''media:subTitle'\''] ! map:merge((@type|@lang|@href) !
map{local-name(): string()})}",
"pubDate": "if (exists(pubDate) and string-length(pubDate) > 0) then pubDate else
fn:current-date()",
"seasonNum": "*[name()='\''xumo:episodic'\'']/*[name()='\''xumo:seasonNum'\'']/text()",
"episodeNum": "*[name()='\''xumo:episodic'\'']/*[name()='\''xumo:episodeNum'\'']/text()",
```

```
"type": "*[name()='\''xumo:episodic'\'']/@type",
"seriesId": "*[name()='\''xumo:episodic'\'']/*[name()='\''xumo:seriesId'\'']/text()",
"rotationStart": "if (exists(dcterms:valid) and string-length(dcterms:valid) > 0 and
contains(dcterms:valid, '\''start='\'')) then (if (contains(dcterms:valid, '\'';'\'')) then
substring-before(substring-after(translate(dcterms:valid, '\'' '\'', '\''\''),
'\''start='\''), '\'';'\'') else substring-after(translate(dcterms:valid, '\'' '\'',
'\''\''), '\''start='\'')) else (fn:current-date())",
"rotationEnd": "if (exists(dcterms:valid) and string-length(dcterms:valid) > 0 and
contains(dcterms:valid, '\''end='\'')) then (if (contains(dcterms:valid, '\'';'\'')) then
substring-before(substring-after(translate(dcterms:valid, '\'' '\'', '\''\''),
'\''end='\''), '\'';'\'') else substring-after(translate(dcterms:valid, '\'' '\'',
'\''\''), '\''end='\'')) else (fn:current-date())",
"rating": "*[name()='\''media:rating'\'']/text()",
"cast": "",
"genres": "if (*[name()='\''xumo:episodic'\''][@type='\''series'\'']) then
*[name()='\''media:keywords'\'']/text() else *[name()='\''media:category'\'']/text()",
"cuePoints": "*[name()='\''xumo:cuePoints'\'']/text()"
},
"provider": "31f5c94081bef786",
"packages": [
"958faece049b6a42"
],
"isPrivate": false
}'
```

Monitor ingestions

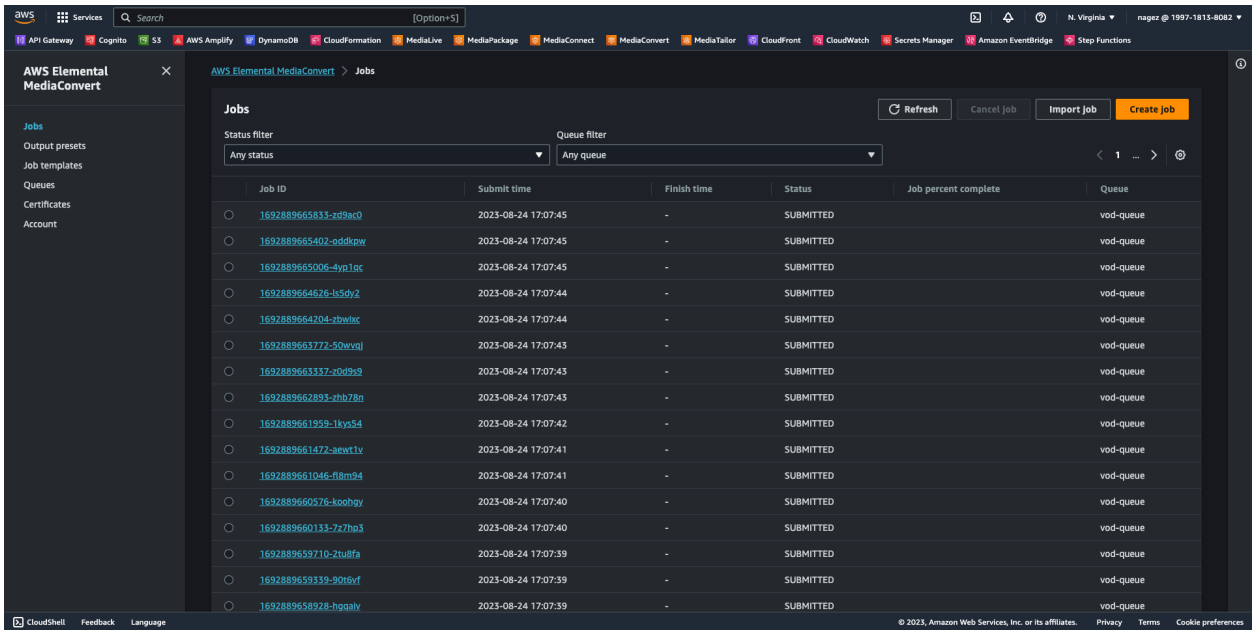


1. Navigate to the “Ingestions” section
2. Click a desired item

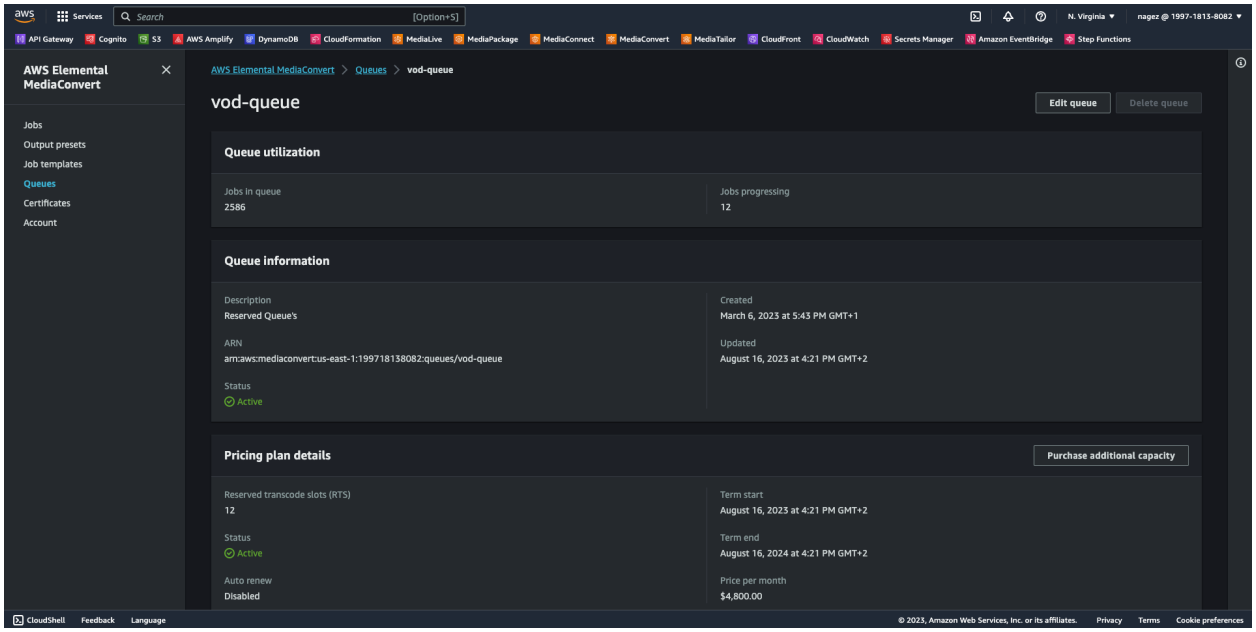


3. A new page opens
4. We have multiple sections, the ingested page above the status, in our case “COMPLETED”, the ingest URL, the Failures, the Saved and the Jobs sections.
    - Failures section, displays ERROR/WARNING logs
    - Saved section, displays saved episodes, movies or series within the Database
    - Jobs section, displays the MediaConvert job status for ingested episodes and/or movies

When an asset is saved in the database, it is sent right away, to MediaConvert for Transcoding.

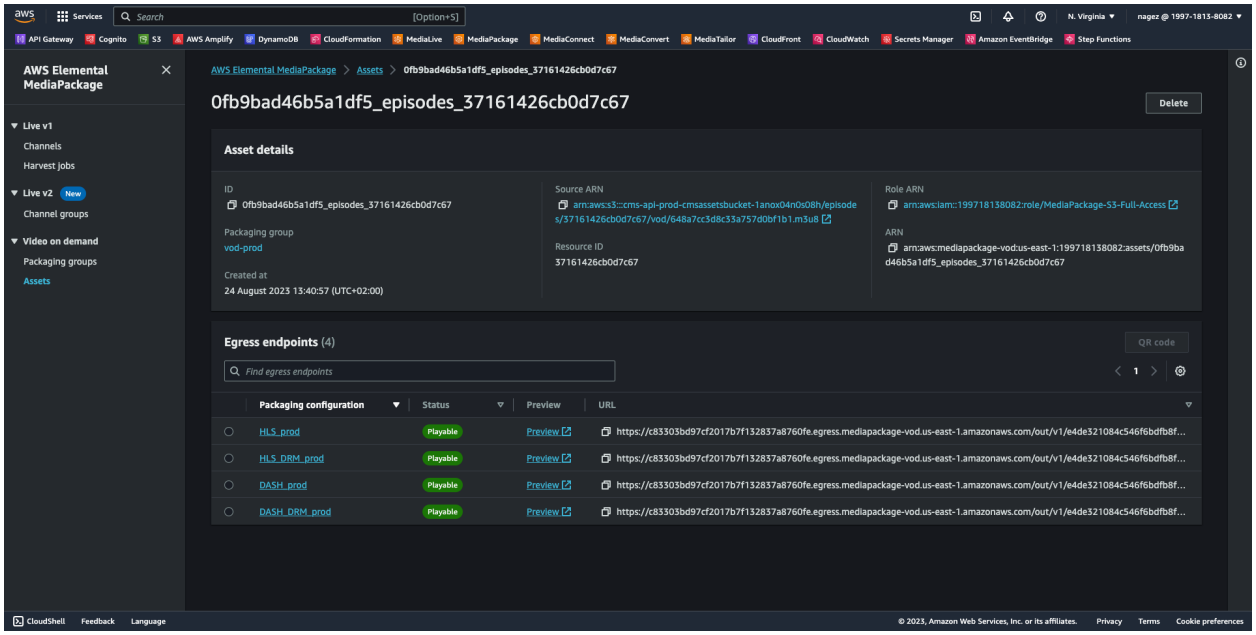


The element is added to the reserved queue.



Once the MediaConvert Job status switches to “**COMPLETE**”, a Lambda function is triggered and a Packaging Job is created in MediaPackage to create HLS with DRM and DASH with and without DRM playlists.

It can be monitored in **MediaPackage** > **Video On Demand Section** > **Assets**



All MediaPackage assets name are formatted the following way:

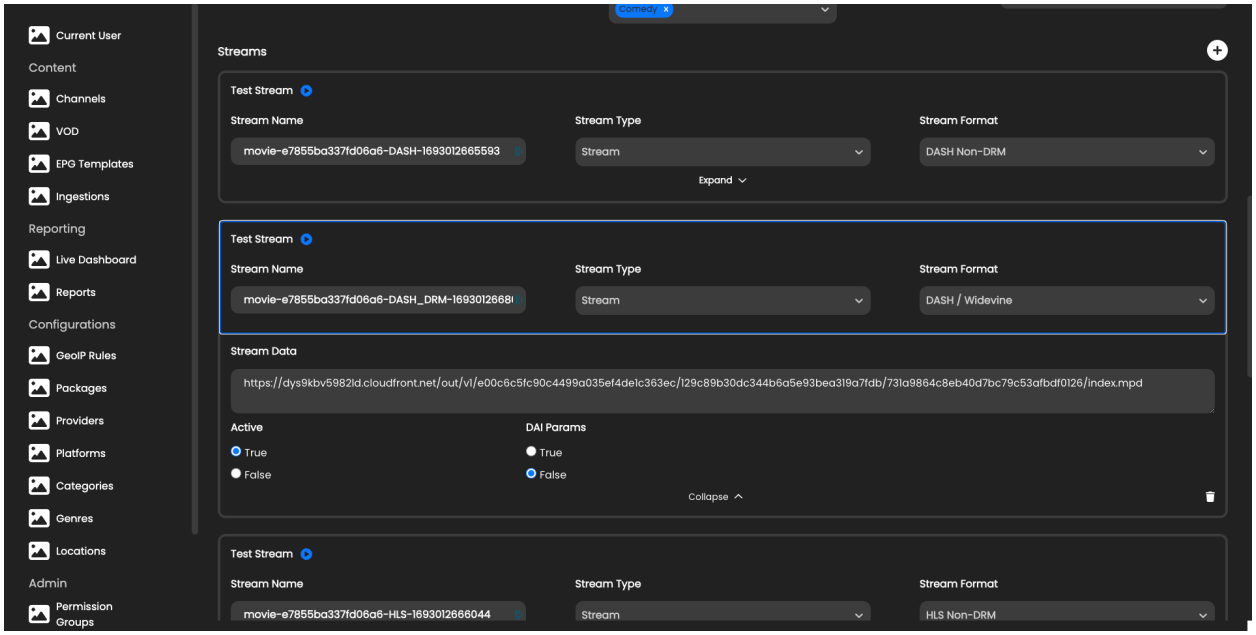
INGESTION-ID_ASSET-TYPE_ASSET-ID

where ASSET-TYPE can be:

- movies or movie
- episodes or episode

In order to playback assets you need to request the manifest using the CloudFront domain since content is protected using CDN Authorization, you can read more here: https://docs.aws.amazon.com/mediapackage/latest/ug/cdn-auth.html

When the MediaPackage Jobs are done, streams are added to the corresponding asset:



Status

- Pages can have 3 status:
  - **COMPLETED**
  - **FAILED**
  - **IN_PROGRESS**
- Jobs can have 9 status
  - **MEDIACONVERT_SUBMITTED**
  - **MEDIACONVERT_PROGRESSING**
  - **MEDIACONVERT_COMPLETE**
  - **MEDIACONVERT_ERROR**
  - **MEDIACONVERT_CANCELED**
  - **MEDIAPACKAGE_PROGRESSING**
  - **MEDIAPACKAGE_ERROR**
  - **MEDIAPACKAGE_COMPLETE**
  - **MEDIAPACKAGE_PLAYABLE**

S3 Location

MediaConvert produces an HLS playlist used by MediaPackage, this HLS playlist is stored within an S3 bucket (one S3 bucket per environment), along with thumbnails.

Movies:

- **HLS path:** cms-api-ENV-cmsassetsbucket-UUID/movies/MOVIE-ID/vod/Filename.m3u8
- **Thumbnails path:** cms-api-ENV-cmsassetsbucket-UUID/movies/MOVIE-ID/

Episodes:

- **HLS path:** cms-api-ENV-cmsassetsbucket-UUID/episodes/EPISODE-ID/vod/Filename.m3u8
- **Thumbnails path:** cms-api-ENV-cmsassetsbucket-UUID/episodes/EPISODE-ID/

Series:

- **HLS path:** cms-api-ENV-cmsassetsbucket-UUID/series/SERIES-ID/vod/Filename.m3u8
- **Thumbnails path:** cms-api-ENV-cmsassetsbucket-UUID/series/SERIES-ID/

